Data Communication & Networking

Data & Signals

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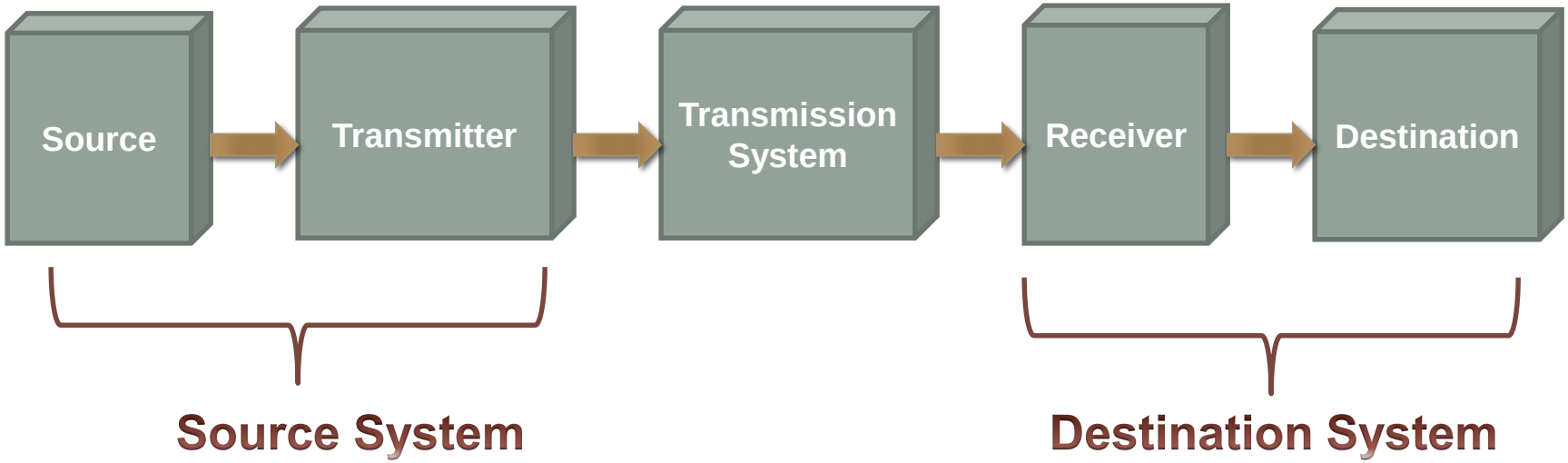
References

- ***Data Communication and Networking*** by Behrouz A. Forouzan
- ***Data and Computer Communication*** by William Stallings

Data Communication System



Data Communication System

- **Source**
  - Generate Data to be transmitted
  - Usually a Computer (can be any other electronic device such as telephone handset, video camera)
- **Transmitter**
  - Data is converted into Signals
  - Use **modem**
- **Communication Medium**
  - Signal is sent to the receiver through a communication medium.
  - Twisted pair of wire, coaxial cable, optical fiber or wireless communication system.

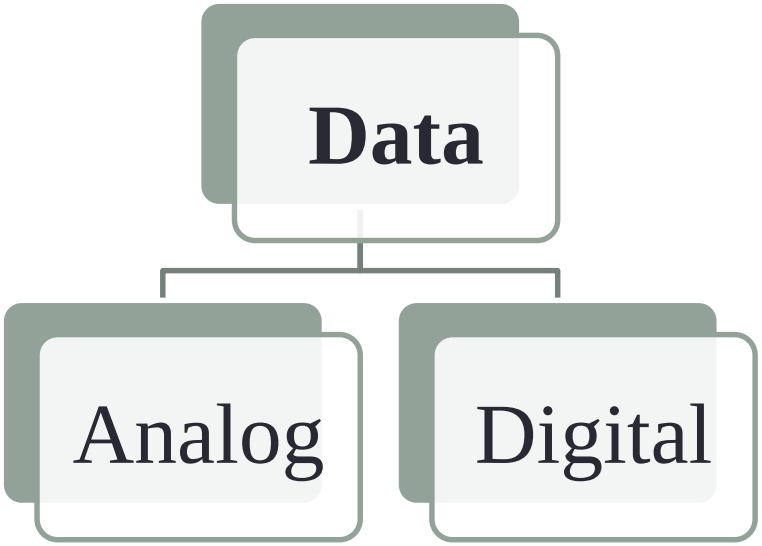
Data Communication System

- **Receiver**
  - Receive the signals and convert it into data or message

- **Destination**
  - Data is absorbed
  - Again there can be a computer system, a telephone, a television etc.

Data

- Information conveys some meaning based on some **mutually-agreed up rules or conventions** between a **sender** and **receiver**.
- Two types of data



Types of Data

- **Analog Data**
  - Representation variable takes on **continuous values** in some interval.
  - E.g. voice, temperature

- **Digital Data**
  - Representation variable takes on **discrete values** in given interval
  - A finite or countable number of values.
  - E.g. text, character string

Signals

- Electrical, electronic or optical representation of data, which can be transmitted through a communication medium.
  - E.g. a microphone converts **voice data** into **voice signal**, which can be sent over a **pair of wire.**

❖ **Analog Signals**

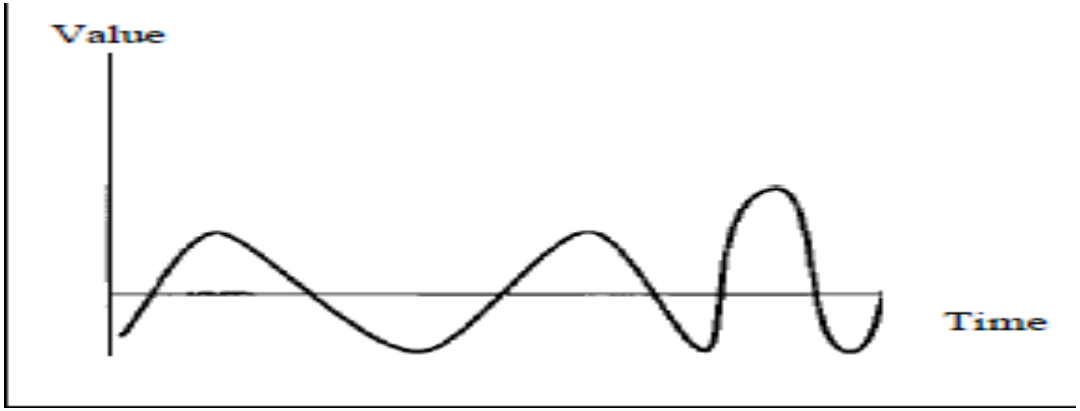
- An infinite number of values in a range.
- Signal intensity varies smoothly over time.

❖ **Digital Signals**

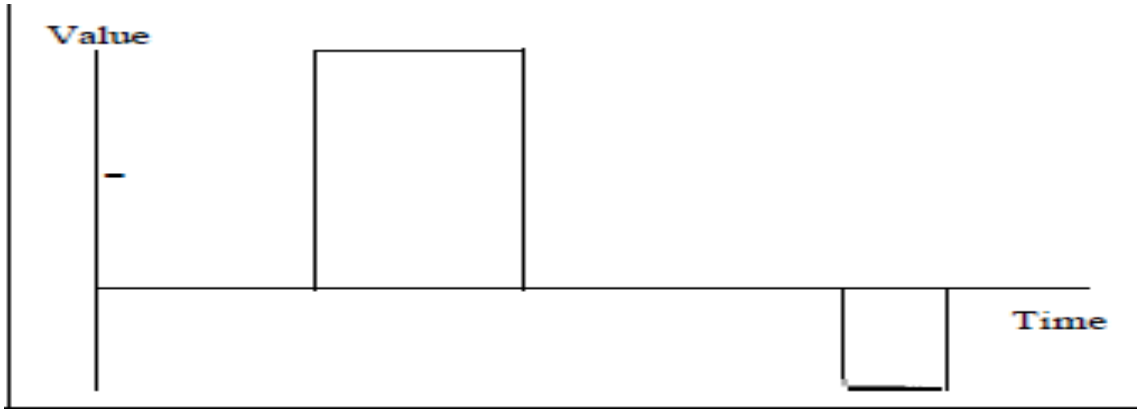
- Can have only a limited number of defined values.
- Usually two values, 0 and 1.

Signals

- Analog Signals



- Digital Signals



Signals

- **Signaling**
  - An act of sending signals over communication medium

- **Transmission**
  - Communication of data by propagation and processing is known as transmission.

Signal Characteristics

- A signal can have both **time-domain** and **frequency domain** representations.

❖ **Time – Domain representation:**

- Signal is represented as a function of time.

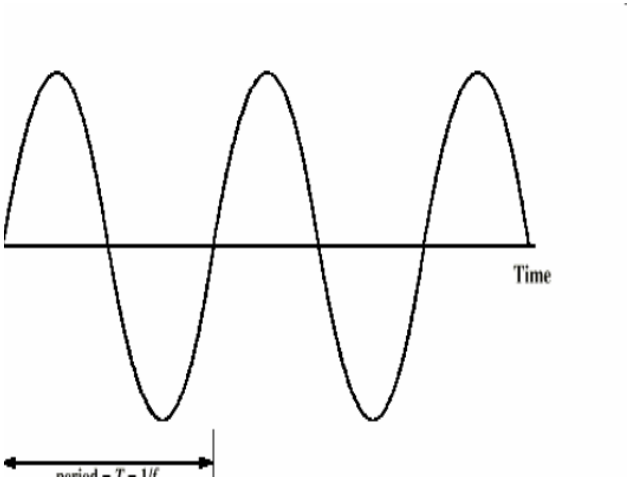
❖ **Frequency – Domain representation:**

- Signal is represented as a function of frequency.
- Signal can be considered as a composition of different frequency components.

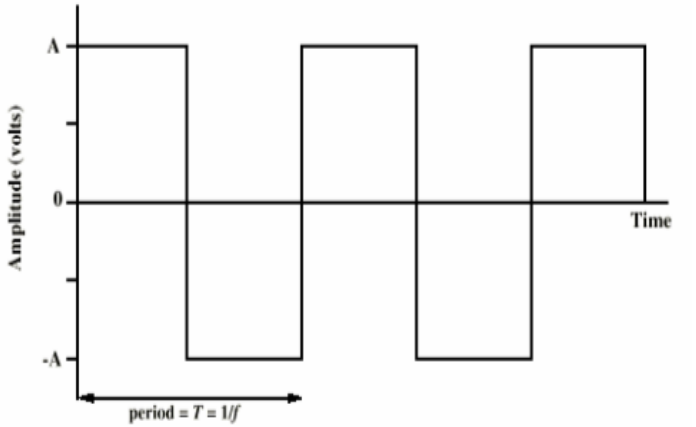
Time Domain Concepts

❖**Periodic Signal**

- A signal that completes a certain pattern within a measurable time frame. (This time is called a **period (T)**)
- Periodic signals can be found in both analog and digital signals.



Periodic Analog Signal



Periodic Digital Signal

Time Domain Concepts

- Three important characteristics of periodic signal:

1. **Amplitude:** Instantaneous value of a signal at any time. This is measured in **volts** (V).

   - **Peak Amplitude** : Absolute value of signal's highest intensity. (Unit - vlots)

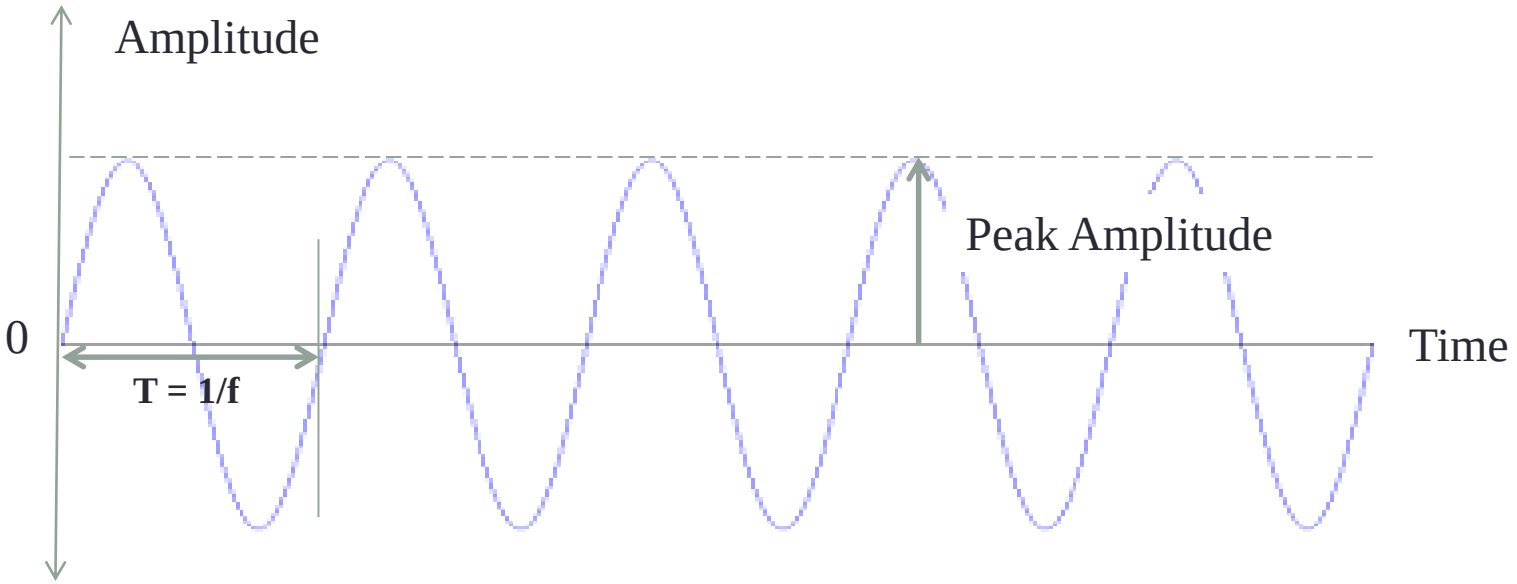
2. **Frequency:** Number of periods in one second. (Inverse of period/ T) Unit is **Hertz** (1Hz = 1/s).

$$\mathrm{f} = \frac{1}{T} \text{ and } T = \frac{1}{f}$$

3. **Phase:** Absolute position of the waveform relative to an arbitrary Origin. This is measured in **degrees** (°) or **radians** (rad)

Time Domain Concepts

- **Sine Wave**



A sine wave can be represented by 3 parameters

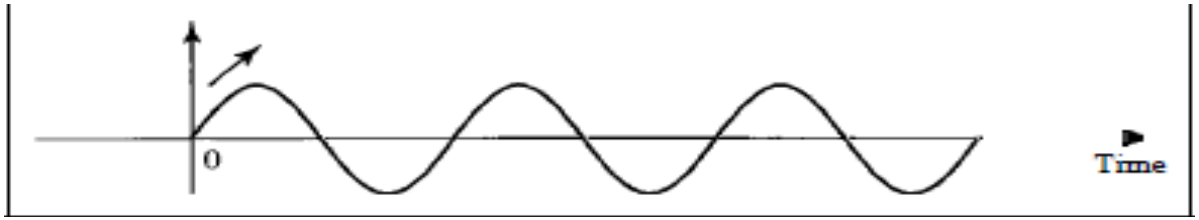
i. Peak Amplitude (A)
ii. Frequency (f)
iii. Phase (**φ**)

$$S(t) = A.\sin(2\pi f t + \varphi)$$

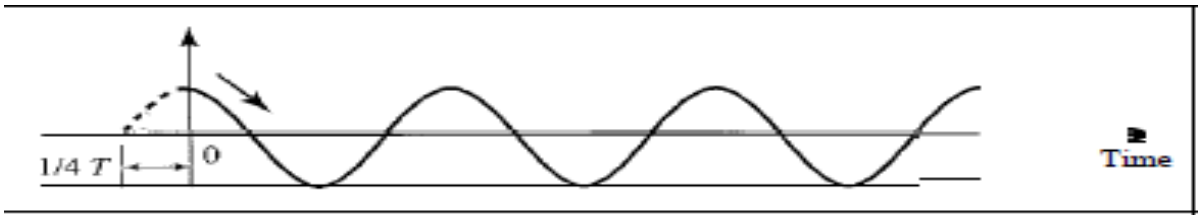
Time Domain Concepts

- Three sine waves with same amplitude and frequency, but different **phases**.

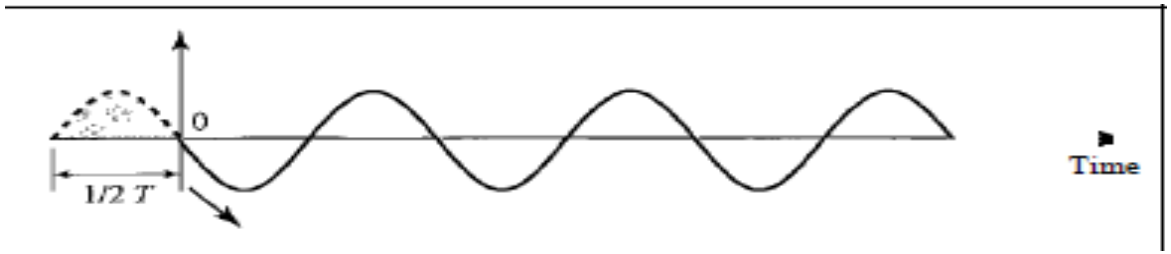
$s(t) = \sin(t)$ φ=0°



$s(t) = \sin(t + 90°)$ φ=90°

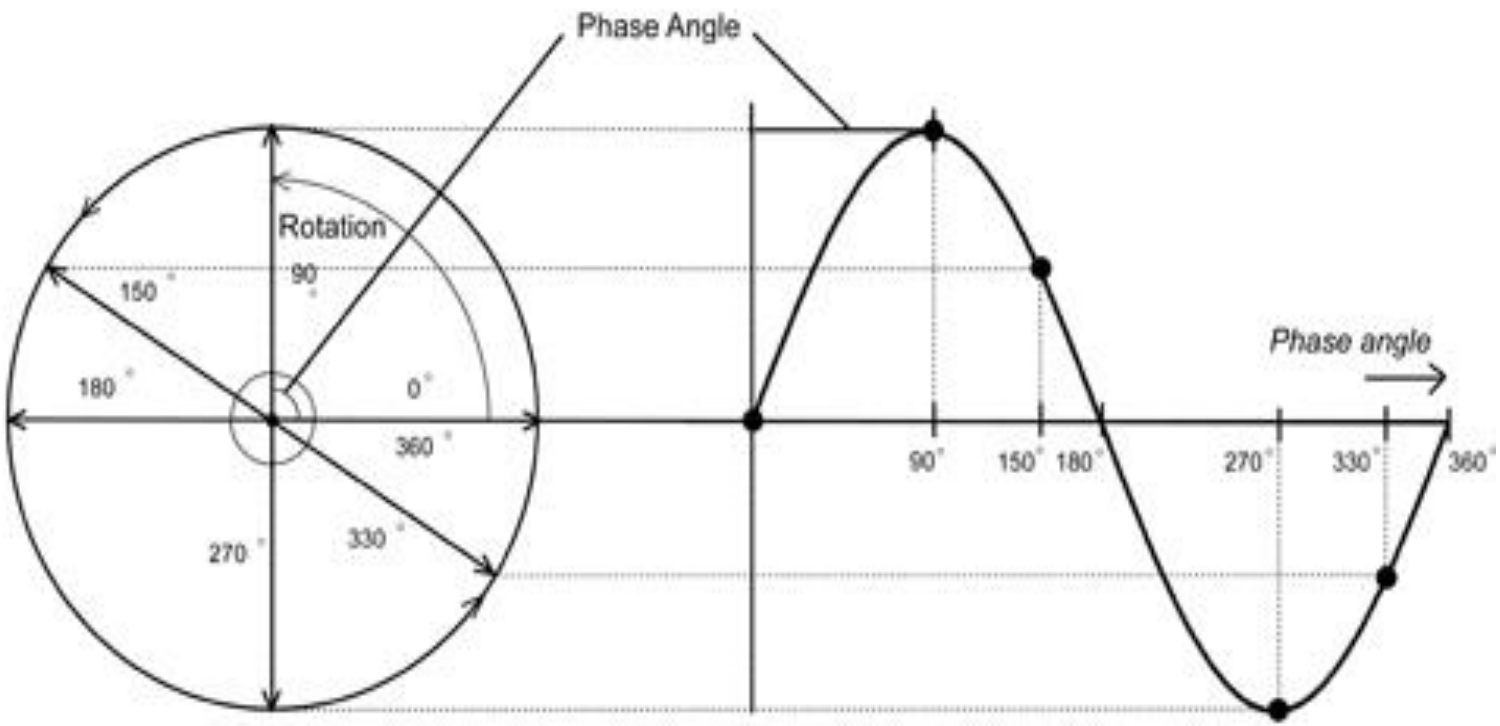


$s(t) = \sin(t + 180°)$ φ=180°



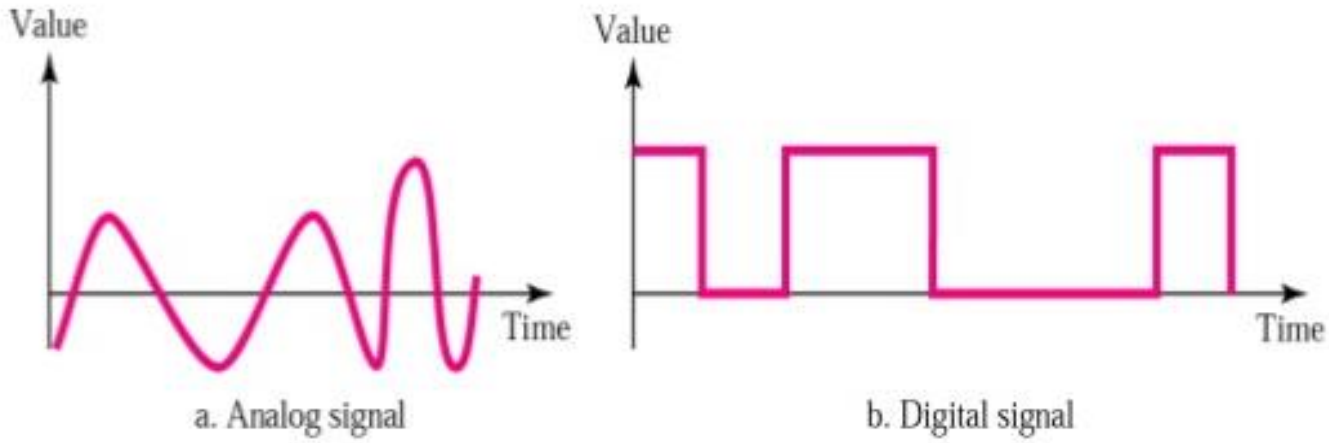
Time Domain Concepts

- **Phase angle and the sine wave**



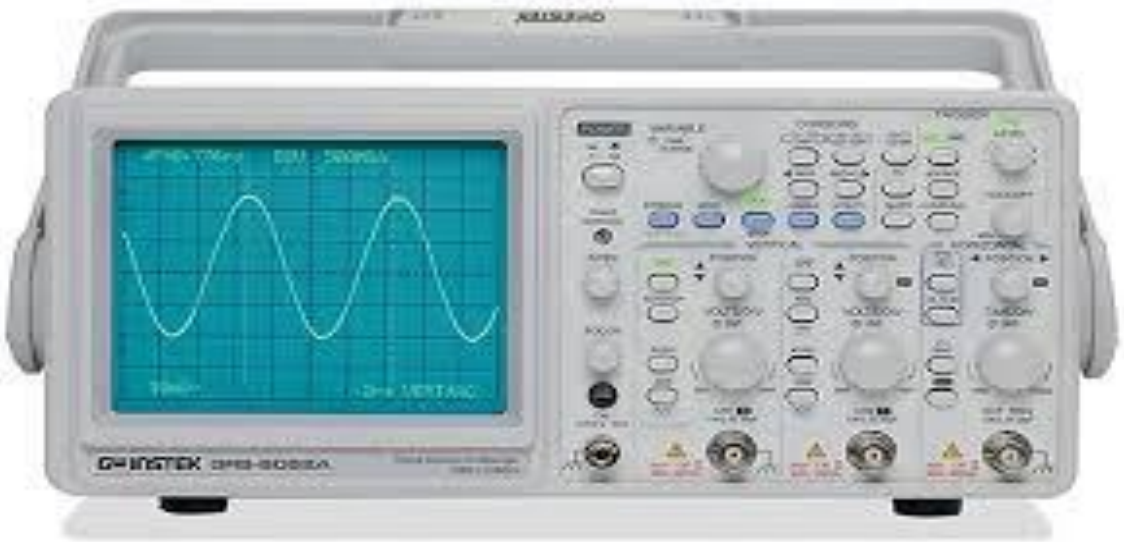
Time Domain Concepts

- **Aperiodic Signals**
  - An **Aperiodic Signal** is a signal that changes **without exhibiting a pattern** or cycle that repeats over time.
  - An Aperiodic Signal can be decomposed into infinite number of periodic signals.



Time Domain Concepts

- Time domain representation displays a signal using the **time-domain plot** which shows changes in signal with respect to the time.
- The time domain plot can be visualized with the help of an **oscilloscope**.

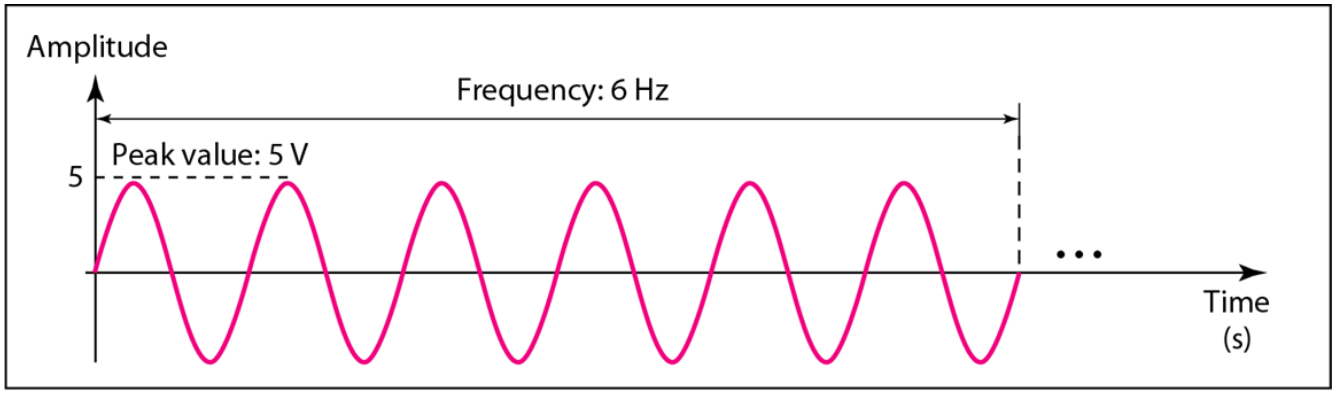


Frequency Domain Concepts

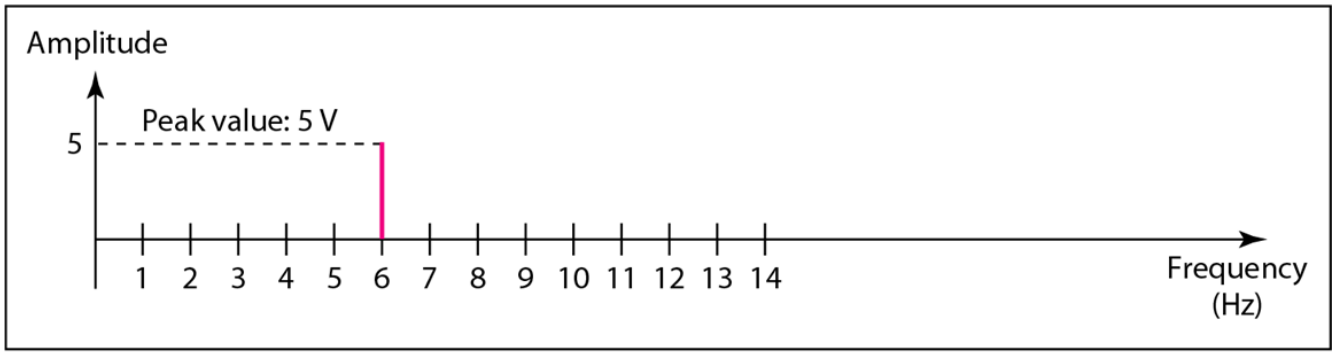
- **Frequency-domain plot** can be used to show the relationship between **amplitude** and **frequency**.
- The above relationship under frequency domain representation can be displayed using **spectrum analyzer**.
- Frequency-domain plot is concerned with only the **peak value** of the amplitude and the **frequency**.

Frequency Domain Concepts

- **Example**



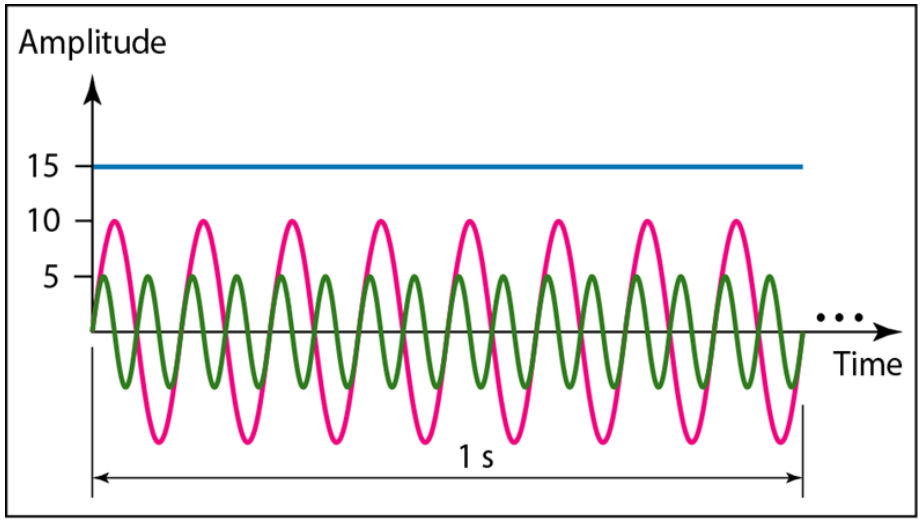
a. A sine wave in the time domain (peak value: 5 V, frequency: 6 Hz)



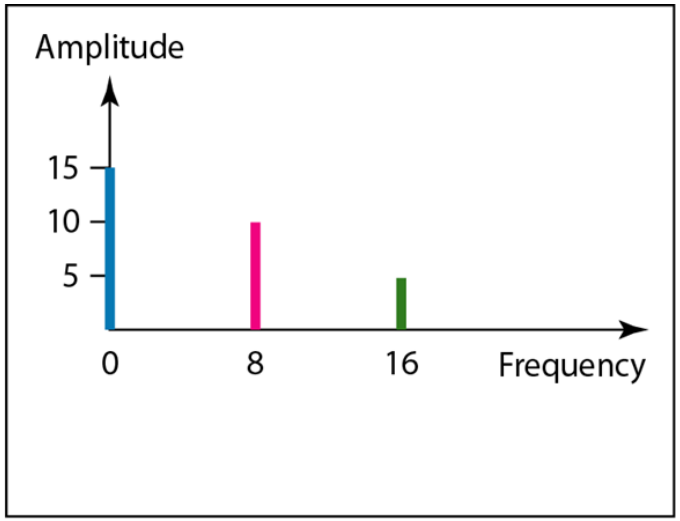
b. The same sine wave in the frequency domain (peak value: 5 V, frequency: 6 Hz)

Frequency Domain Concepts

- Frequency domain is more compact and useful when dealing with more than one sine wave.



a. Time-domain representation of three sine waves with frequencies 0, 8, and 16



b. Frequency-domain representation of the same three signals

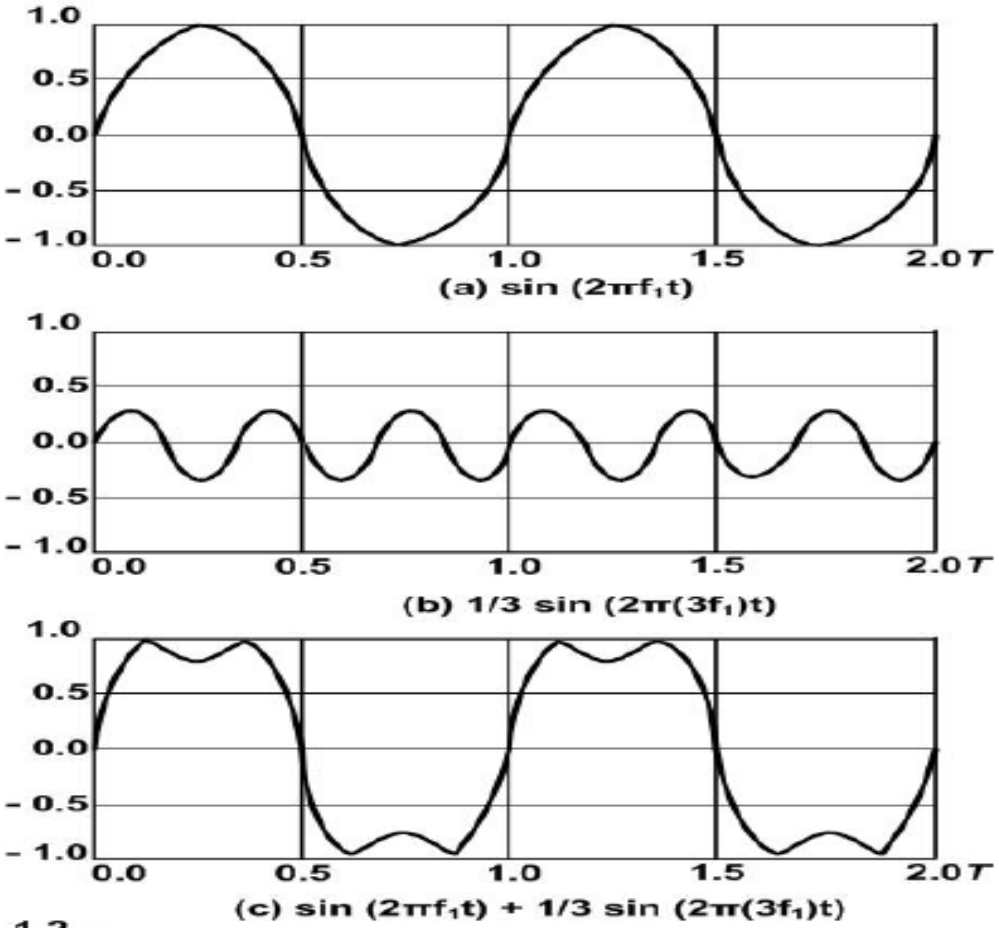
Frequency Domain Concepts

- **Composite Signals**
  - A Composite signal is made of many simple sine waves.
  - Composite signals are used in data communication.
  - According to **Fourier Analysis**, any composite signal can be represented by a combination of sine waves. E.g.

$$\mathrm{S(t)} = \sin(\omega t) + \frac{1}{3}\sin(\omega t) \text{ where } \omega = 2\pi f$$

  - The frequency domain function, $s(f)$, specifies the constituent frequencies of the signal.

Frequency Domain Concepts



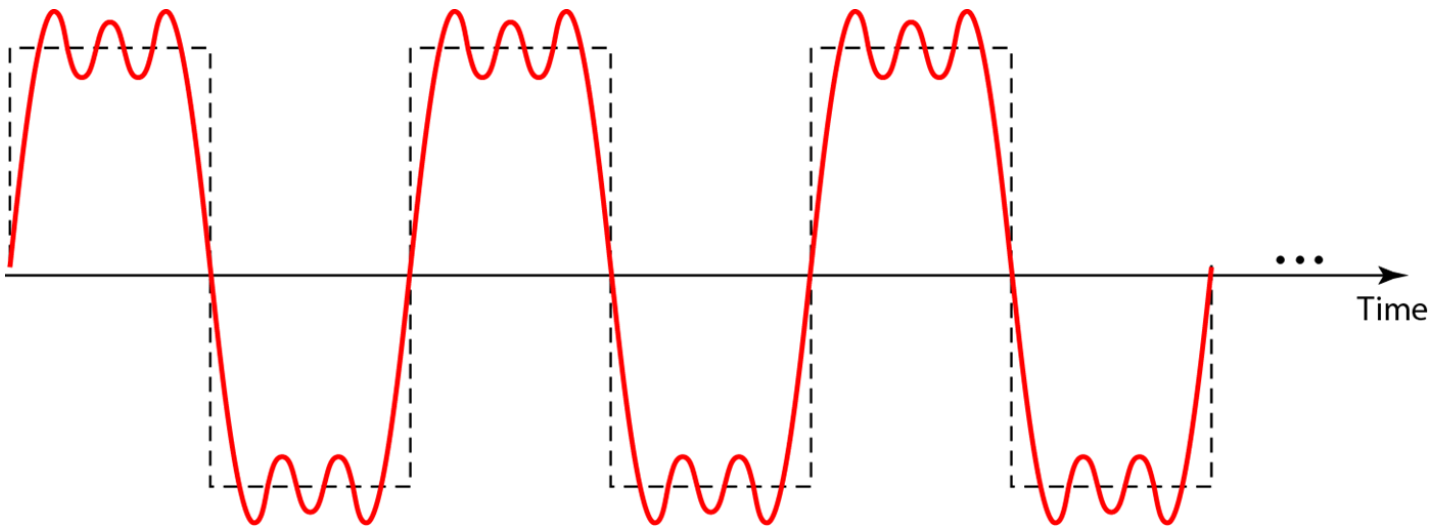
Frequency Domain Concepts

- **Composite Signals**
  - Composite Signals can be **periodic** or **aperiodic**
  - **Frequency Spectrum :** Range of frequencies that a signal contains.
  - **Bandwidth** : Range of frequencies over which most of the signal energy of the signal is contained. **(Effective Bandwidth)**

    For a composite signal, it is the difference between the highest and the lowest frequencies contained in that signal **(Absolute Bandwidth)**

Frequency Domain Concepts

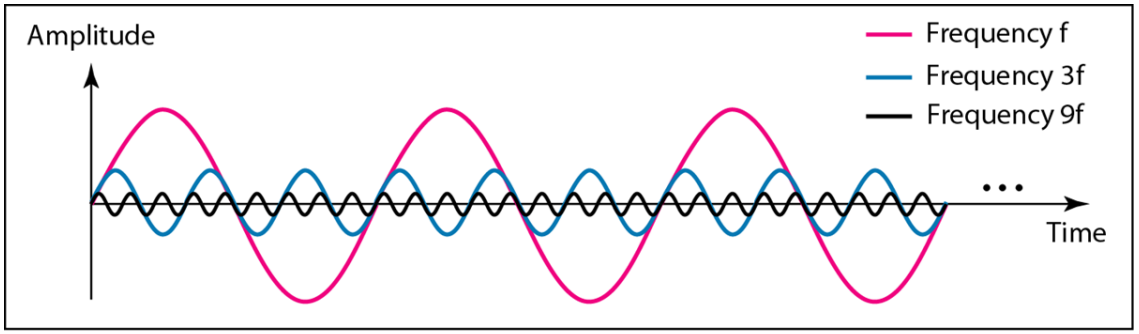
- **Composite Signals**



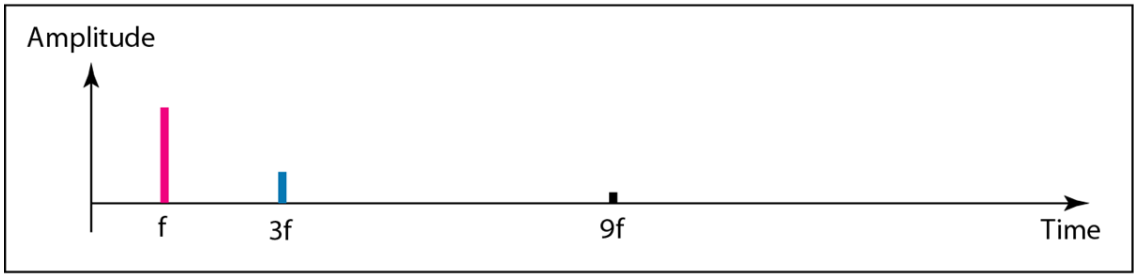
Original Composite Signal

Frequency Domain Concepts

- **Composite Signals**



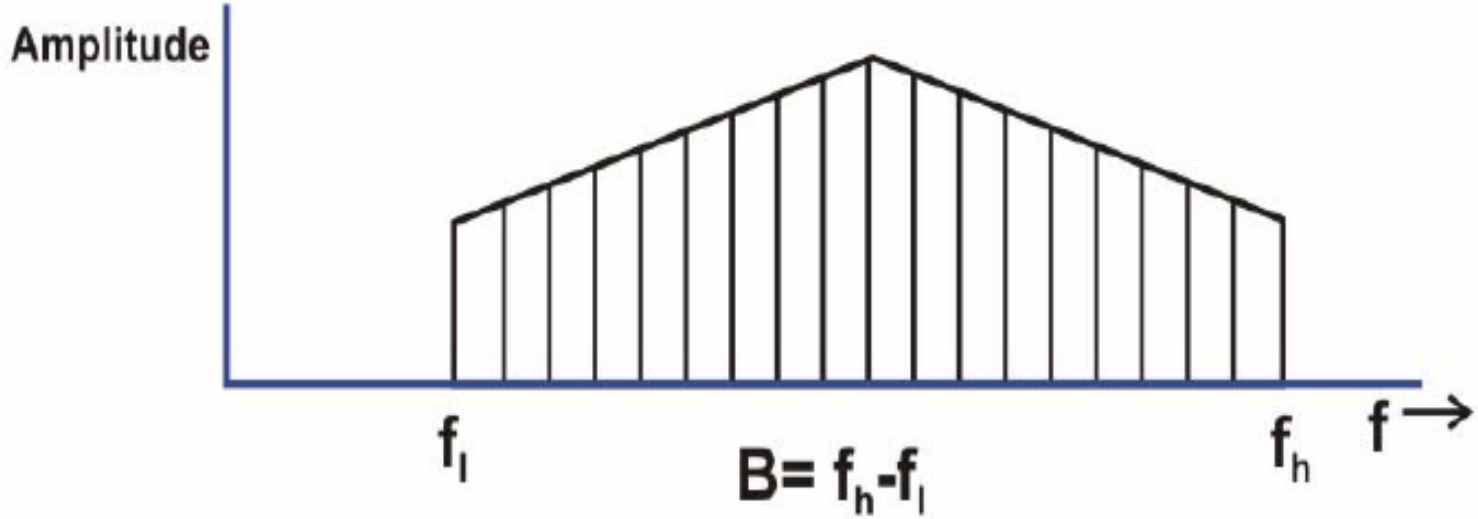
a. Time-domain decomposition of a composite signal



b. Frequency-domain decomposition of the composite signal

Decomposition of Composite Signal

Frequency Domain Concepts



Bandwidth of a composite Signal